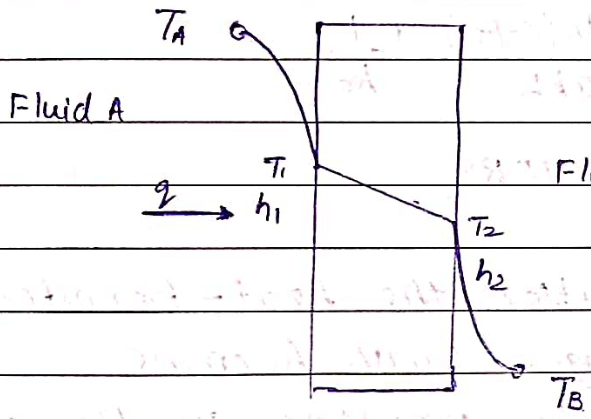
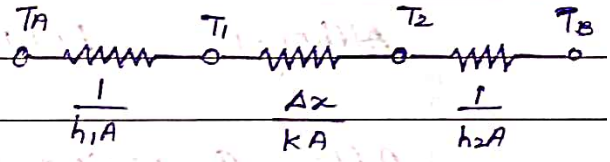
HEAT EXCHANGERS :

Overall HEAT-TRANSFER COEFFICIENT

We have already discussed that heat transfer through the plane wall is expressed as

$$q = \frac{T_A - T_B}{1/h_1A + \Delta x/kA + 1/h_2A}$$



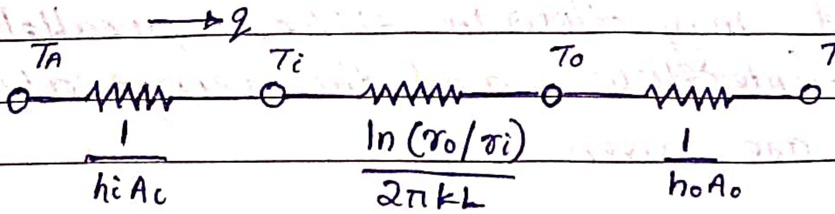


Where T_A and T_B are the fluid temperatures on each side of the wall. The overall heat transfer Coefficient U is defined by the relation.

$$q = UA\Delta T_{overall}$$

In application one fluid flows on the inside of the smaller tube while the other fluid flows in the annular space between the two tubes.

$$q = \frac{T_A - T_B}{\frac{1}{h_iA_i} + \frac{\ln(r_o/r_i)}{2\pi kL} + \frac{1}{h_oA_o}}$$



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where i and o represents inside and outside of the smaller inner tube. The overall heat-transfer coefficient may be based on either the inside or outside area of the tube at the discretion of the designer. Accordingly

$$U_i = \frac{1}{\frac{1}{h_i} + \frac{A_i \ln(r_o/r_i)}{2\pi k L} + \frac{A_i}{A_o}\frac{1}{h_o}}$$

$$U_o = \frac{1}{\frac{A_o}{A_i}\cdot\frac{1}{h_i} + \frac{A_o \ln(r_o/r_i)}{2\pi k L} + \frac{1}{h_o}}$$

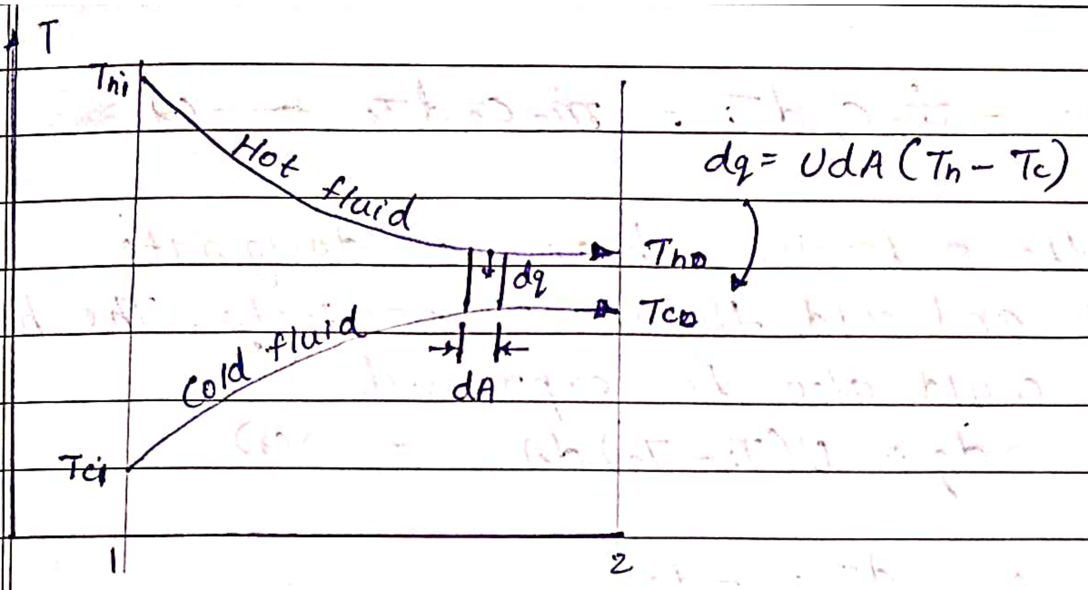
~~Although~~ FOULING FACTORS

After a period of operation the heat-transfer surfaces for a heat exchanger may become coated with various deposits present in the flow systems, or the surfaces may become corroded as a result of the interaction between the fluids & the material used for construction of the heat exchanger. The coating represents an additional resistance to the heat flow & results in decreased performance. The overall effect is usually represented by a fouling factor or fouling resistance R_f

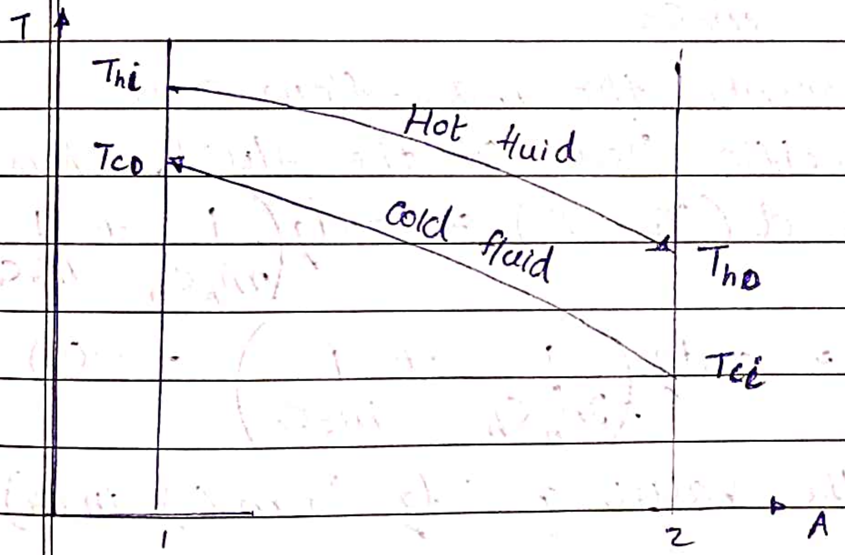
~~$R_f =$~~

THE LOG MEAN TEMPERATURE DIFFERENCE

The fluids may flow in either parallel flow or counterflow, and the temperature profiles are given as



Parallel flow



$$q = UA\Delta T_m \quad \text{——— (1)}$$

where

U = overall heat-transfer coefficient

A = surface area for heat transfer consistent with definition of U

ΔT_m = suitable mean temperature difference across heat exchanger.

For a parallel-flow heat exchanger, the heat transferred through an element of area dA may be written

$$dq = -\dot{m}_h C_h \, dT_h = \dot{m}_c C_c \, dT_c \quad \text{——} (2)$$

where the subscripts h and c designate the hot and cold fluids, respectively. The heat transfer could also be expressed

$$dq = U(T_h - T_c)\, dA \quad \text{——} (3)$$

From (2) $dT_h = \frac{-dq}{\dot{m}_h C_h}$

and $dT_c = \frac{dq}{\dot{m}_c C_c}$

where $\dot{m}$ represents the mass-flow rate and c is the specific heat of the fluid. Thus

$$dT_h - dT_c = d(T_h - T_c) = -dq\left(\frac{1}{\dot{m}_h C_h} + \frac{1}{\dot{m}_c C_c}\right)$$

$$d(T_h - T_c) = -dq\left(\frac{1}{\dot{m}_h C_h} + \frac{1}{m_c C_c}\right) \quad \text{——} (4)$$

substituting the value of dq from (3) in (4)

$$d(T_h - T_c) = -U(T_h - T_c)\left[\frac{1}{\dot{m}_h C_h} + \frac{1}{m_c C_c}\right] dA$$

$$\frac{d(T_h - T_c)}{T_h - T_c} = -U\left(\frac{1}{\dot{m}_h C_h} + \frac{1}{\dot{m}_c C_c}\right) dA \quad \text{——} (5)$$

On integrating both sides with suitable limits

$$\ln\frac{(T_{ho} - T_{co})}{(T_{hi} - T_{ci})} = -UA\left(\frac{1}{\dot{m}_h C_h} + \frac{1}{\dot{m}_c C_c}\right)$$

—— (5)

From equation (2)

$$\dot{m}_h c_h = \frac{q}{T_{hi} - T_{ho}} \quad \text{and} \quad \dot{m}_c c_c = \frac{q}{T_{c_o} - T_{ci}}$$

On substituting the above values in (5) we get

$$\ln \frac{(T_{ho} - T_{co})}{(T_{hi} - T_{ci})} = -UA\left(\frac{T_{hi} - T_{ho}}{q} + \frac{T_{co} - T_{ci}}{q}\right)$$

$$\ln \frac{(T_{ho} - T_{co})}{T_{hi} - T_{ci}} = \frac{-UA}{q}\left[T_{hi} - T_{ci} - (T_{ho} - T_{co})\right]$$

$$\text{or} \quad q = \frac{UA\left[(T_{ho} - T_{co}) - (T_{hi} - T_{ci})\right]}{\ln\left[\frac{T_{ho} - T_{co}}{T_{hi} - T_{ci}}\right]}$$

Hence the mean temperature difference

$$\text{as } \Delta T_m = \frac{(T_{ho} - T_{co}) - (T_{hi} - T_{ci})}{\ln\left[(T_{ho} - T_{co})/(T_{hi} - T_{ci})\right]} \quad \text{—— (7)}$$

This temperature difference is called the log mean temperature difference (LMTD).

We know from equation (5) that

$$\frac{d(T_h - T_c)}{T_h - T_c} = -U\left(\frac{1}{\dot{m}_h c_h} + \frac{1}{\dot{m}_c c_c}\right) dA$$

$$\& \ dq = m_h c_h \, dT_h = -\dot{m}_c c_c \, dT_c$$

Integrating both sides with suitable limits θ_1 and θ_2 where $\theta_1 = T_{hi} - T_{co}$ and $\theta_2 = T_{ho} - T_{ci}$

and substituting $\dot{m}_h c_h = \frac{q}{T_{hi} - T_{ho}}$ and

$\dot{m}_c c_c = \frac{-q}{T \ldots - T \ldots}$.

$$\int_{\theta_1}^{\theta_2} \frac{d(T_h - T_c)}{(T_h - T_c)} = \int -U\left[\frac{T_{hi} - T_{ho}}{q} + \frac{T_{ci} - T_{co}}{q}\right] dA$$

$$\Rightarrow \left[\ln (T_h - T_c)\right]_{\theta_1}^{\theta_2} = -\frac{U}{q}\left[T_{hi} - T_{co} - (T_{ho} - T_{ci})\right]$$

$$= \ln\left(\frac{T_{ho} - T_{ci}}{T_{hi} - T_{co}}\right) = \frac{UA}{q}\left[T_{ho} - T_{ci} - (T_{hi} - T_{co})\right]$$

$$\therefore q = UA \frac{\left[(T_{ho} - T_{ci}) - (T_{hi} - T_{co})\right]}{\ln\left[\frac{T_{ho} - T_{ci}}{T_{hi} - T_{co}}\right]}$$

Assumptions for LMTD involves

(1) the fluid specific heats do not vary with temperature

(2) the convection heat-transfer coefficients are constant throught the heat exchanger

If a heat exchanger other than the double-pipe type is used, the heat transfer is calculated by using a correction factor applied to the LMTD for a counterflow double-pipe arrangement with the same hot and cold fluid temperatures. The heat transfer equation then takes the form

$$q = UAF\Delta T_m$$

$F = 1$ for boiling or condensation

EFFECTIVENESS -NTU METHOD

The effectiveness method offers many advantages for analysis of problems in which a comparison between various types of heat exchangers must be made for purposes of selecting the type best suited to accomplish a particular heat-transfer objective

We define the heat exchanger effectiveness as

$$\text{Effectiveness } \epsilon = \frac{\text{actual heat transfer}}{\text{max. possible heat transfer}}$$

Actual heat transfer may be computed by calculating either the energy lost by the hot fluid or the energy gained by the cold fluid.

$$q = \dot{m}_h c_h (T_{hi} - T_{ho}) = \dot{m}_c c_c (T_{co} - T_{ci})$$

To determine the maximum possible heat transfer for the exchanger, we first recognize that this maximum value could be attaine a temp. change equal to the maximum temperatures for hot & cold fluids. The fluid that undergoes max. temp. difference is one with "minimum" value of $\dot{m}c$

$$q_{max} = (\dot{m}c)_{min} (T_{hi} - T_{ci})$$

The minimum fluid may be either the hot or cold fluid, depending on the mass flow rates & specific heats

$$\epsilon_h = \frac{\dot{m}_h c_h (T_{hi} - T_{ho})}{\dot{m}_h c_h (T_{hi} - T_{ci})} = \frac{T_{hi} - T_{ho}}{T_{hi} - T_{ci}} \quad \text{——— (1a)}$$

$$\epsilon_c = \frac{\dot{m}_c c_c (T_{co} - T_{ci})}{\dot{m}_c c_c (T_{hi} - T_{ci})} = \frac{T_{co} - T_{ci}}{T_{hi} - T_{ci}} \quad \text{——— (1b)}$$

In a general way effectiveness is expressed as

$$\epsilon = \frac{\Delta T \text{ (minimum fluid)}}{\text{Maximum temperature diff. in HE}}$$

Writing effectiveness in parallel flow HE,

$$\ln\left[\frac{T_{ho} - T_{co}}{T_{hi} - T_{ci}}\right] = -UA\left[\frac{1}{m_h c_h} + \frac{1}{m_c c_c}\right]$$

or $$\ln\left[\frac{T_{ho} - T_{co}}{T_{hi} - T_{ci}}\right] = -UA\left[\frac{1}{C_h} + \frac{1}{C_c}\right]$$

$$\therefore \frac{T_{ho} - T_{co}}{T_{hi} - T_{ci}} = e^{-UA[1/C_h + 1/C_c]} \quad \text{--- (2)}$$

We know that the ~~temperature~~ energy lost by the hot fluid is equal to the energy gained by the cold fluid

$$m_h c_h (T_{hi} - T_{ho}) = m_c c_c (T_{co} - T_{ci})$$

or $$C_h (T_{hi} - T_{ho}) = C_c (T_{co} - T_{ci})$$

On rearranging

$$T_{ho} = T_{hi} - \frac{C_c}{C_h}(T_{co} - T_{ci})$$

Substituting the value of T_{ho} from above equation in (2) we get

$$\frac{T_{hi} - \frac{C_c}{C_h}(T_{co} - T_{ci}) - T_{co}}{T_{hi} - T_{ci}} = e^{-UA\left[\frac{1}{C_h} + \frac{1}{C_c}\right]}$$

$$\frac{T_{hi} - \frac{C_c T_{co}}{C_h} + \frac{C_c T_{ci}}{C_h} - T_{co}}{T_{hi} - T_{ci}} = e^{-UA\left[\frac{1}{C_h} + \frac{1}{C_c}\right]}$$

Adding and subtracting T_{ci} ~~for~~ on numerator

$$\cancel{T_{hi}} + \cancel{T} - \cancel{T} + \frac{\cancel{C_c}}{C_h}\cancel{T} - \frac{\cancel{C_c}}{C_h}(\cancel{T_{co}}) + \cancel{T}$$

$$\frac{T_{hi} - T_{ci} + T_{ci} + \frac{C_c}{C_h}T_{ci} - T_{co} - \frac{C_c}{C_h}T_{co}}{T_{hi} - T_{ci}} = e^{-UA\left[\frac{1}{C_h}+\frac{1}{C_c}\right]}$$

$$\Rightarrow \frac{T_{hi} - T_{ci} + T_{ci}\left[1+\frac{C_c}{C_h}\right] - T_{co}\left[1+\frac{C_c}{C_h}\right]}{T_{hi} - T_{ci}} = e^{-UA\left[\frac{1}{C_h}+\frac{1}{C_c}\right]}$$

$$\Rightarrow 1 + \frac{(T_{ci} - T_{co})}{(T_{hi} - T_{ci})}\left[1+\frac{C_c}{C_h}\right] = e^{-UA\left[\frac{1}{C_h}+\frac{1}{C_c}\right]}$$

$$\text{or} \quad 1 - \frac{(T_{co} - T_{ci})}{(T_{hi} - T_{ci})}\left[1+\frac{C_c}{C_h}\right] = e^{-UA\left[\frac{1}{C_h}+\frac{1}{C_c}\right]} \quad \text{——(3)}$$

We know effectiveness $\epsilon = \frac{Q}{Q_{max}}$

$$\epsilon = \frac{C_c(T_{co} - T_{ci})}{C_{min}(T_{hi} - T_{ci})}$$

$$\therefore \frac{T_{co} - T_{ci}}{T_{hi} - T_{ci}} = \frac{C_{min}}{C_c} \cdot \epsilon$$

Substituting the value obtained in (3)

$$1 - \frac{C_{min}}{C_c}\epsilon\left[1+\frac{C_c}{C_h}\right] = e^{-UA\left[\frac{1}{C_h}+\frac{1}{C_c}\right]}$$

$$\therefore \frac{C_{min}}{C_c}\epsilon\left[1+\frac{C_c}{C_h}\right] = 1 - e^{-UA\left[\frac{1}{C_h}+\frac{1}{C_c}\right]}$$

$$\epsilon = \frac{1 - e^{-UA\left[\frac{1}{C_h}+\frac{1}{C_c}\right]}}{\frac{C_{min}}{C_c}\left[1+\frac{C_c}{C_h}\right]}$$

$$= \frac{1 - e^{-UA\left[\frac{1}{C_h}+\frac{1}{C_c}\right]}}{\frac{C_{min}}{C_c} + \frac{C_{min}}{C_h}}$$

Multiplying the num & deno of the exponential term we get

$$\epsilon = \frac{1 - e^{\frac{-UA}{C_{min}}\left[\frac{C_{min}}{C_h} + \frac{C_{min}}{C_c}\right]}}{\left[\frac{C_{min}}{C_c} + \frac{C_{min}}{C_h}\right]}$$

number of heat transfer units (NTU)

$$N = UA / C_{min}$$

$$\therefore \epsilon = \frac{1 - e^{-N\left[\frac{C_{min}}{C_h} + \frac{C_{min}}{C_c}\right]}}{\left[\frac{C_{min}}{C_c} + \frac{C_{min}}{C_h}\right]}$$

Let $C = \frac{C_{min}}{C_{max}}$

Out of C_h and C_c one will be maximum and the other will be minimum

$$\epsilon = \frac{1 - e^{-N[1+C]}}{[1+C]}$$

Significance of NTU

The NTU method is utilized in order to optimize the number of transferred units in the process of exchange of heat. This method is found to be applied in the case of insufficient data gathered from the calculation of LMTD. Helpful in case of unavailability of specific temperatures

- Utilized in the evaluation of rate of mass exchange.
- Simple way to build the relation b/t heat and mass transfer performance & the geometric, physical and operating conditions of heat & mass exchangers.

10) Distinguishing factors b/t LMTD & NTU methods

In accordance with the LMTD method, it is found that this method is considered to be convenient for determination of overall amount of heat transfer.
In NTU method, it is optimized that this method is found to be convenient for predicting the temp. of the outlet of the concerned fluid
NTU method is considered more acceptable as compared to LMTD method.
In NTU method, a "large variety of heat exchange configuration is taken into considerations"

TUTORIAL PROBLEMS

2) Water is heated at a rate of 0.8 kg/s
water gets heated from 30°c to 80°c

Hence $T_{ci} = 30°c$
$T_{co} = 80°c$

Hot oil enters at 120°c and leaves at 85°c

$\therefore$ $T_{hi} = 120°c$
$T_{ho} = 85°c$

Overall heat transfer coeff $U = 125\, w/m^2k$

(a) Heat exchanger in parallel flow arranged
total heat transfer is determined from the energy absorbed by the water.

$$q = m_w C_w \Delta T_w = 0.8 \times 4178 \times [80-30]$$
$$= 167120\, J/s$$
or

since all fluid temp. are known, the LMTD can be calculated

LMTD $\Delta T_{lm} = \dfrac{\theta_1 - \theta_2}{\ln(\theta_1/\theta_2)}$

For parallel flow heat exchanger arrangement

$\theta_1 = T_{hi} - T_{ci}$

$\theta_2 = T_{ho} - T_{co}$

$\theta_1 = 120 - 30 = 90$

$\theta_2 = 85 - 80 = 5$

$\therefore$ LMTD $\Delta T_{lm} = \dfrac{90-5}{\ln(90/5)} = \dfrac{85}{\ln(18)} = 29.40^\circ C$

Since $q = UA(LMTD)$

$A = \dfrac{q}{U(LMTD)} = \dfrac{167120}{125 \times 29.40}$

$= 45.47\ m^2$

(b) When heat exchanger is counter flow arranged

$LMTD = \dfrac{\theta_1 - \theta_2}{\ln(\theta_1/\theta_2)}$

For counter flow heat exchanger arrangement

$\theta_1 = T_{hi} - T_{co}$

$\theta_2 = T_{ho} - T_{ci}$

$\theta_1 = 120 - 80 = 40$

$\theta_2 = 85 - 30 = 55$

$\therefore LMTD = \dfrac{\theta_1 - \theta_2}{\ln(\theta_1/\theta_2)} = \dfrac{40-55}{\ln(40/55)} = 47.10^\circ C$

Since $q = UA(LMTD)$

$A = \dfrac{q}{U(LMTD)} = \dfrac{167120}{125 \times 47.10} = \underline{\underline{28.38\ m^2}}$

(c) When heat exchanger is of shell & tube arranged with two shell pass & four tube passes

$LMTD = \dfrac{\theta_1 - \theta_2}{\ln(\theta_1/\theta_2)}$

where $\theta_1 = T_{hi} - T_{co}$

$\theta_2 = T_{ho} - T_{ci}$

$\theta_1 = 120 - 80 = 40$

$\theta_2 = 85 - 30 = 55$

$$LMTD = \frac{\theta_1 - \theta_2}{\ln(\theta_1/\theta_2)} = \frac{40-55}{\ln(40/55)} = \underline{47.10^\circ C}$$

We have p

$$P = \frac{T_{co} - T_{ci}}{T_{hi} - T_{ci}} = \frac{80-30}{120-30} = 0.55$$

$$R = \frac{T_{hi} - T_{ho}}{T_{co} - T_{ci}} = \frac{120-85}{80-30} = 0.7$$

$F = 0.97$ (correction factor from data book Pg 171)

$$\therefore \quad A = \frac{q}{U(LMTD)\cdot F} = \frac{167120}{125 \times 47.10 \times 0.97} = \underline{29.27\,m^2}$$

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3) Total heat transfer may be obtained from energy balance on the steam

$$q = m_s C_s \Delta T_s = 5.2 \times 1.86 \times [T_{hi} - T_{ho}]$$
$$= 5.2 \times 1.86\,[130 - 110] = 193.44\,kW$$

where m = mass flow rate of steam = ~~1.86 kJ/kg~~ 5.2 kg/s

C = specific heat = 1.86 kJ/kg °C

$T_{hi} = 130 \qquad T_{ho} = 110$

$T_{ci} = 15 \qquad T_{co} = 85$

$$\therefore LMTD = \frac{\theta_1 - \theta_2}{\ln(\theta_1/\theta_2)} \qquad \text{where } \theta_1 = T_{hi} - T_{co}, \quad \theta_2 = T_{ho} - T_{ci}$$

$\therefore \theta_1 = 130 - 85 = 45 \qquad \theta_2 = 110 - 15 = 95$

$$\therefore LMTD = \frac{45-95}{\ln(45/95)} = 66.91^\circ C$$

Capacity ratio $R = \dfrac{T_{hi} - T_{ho}}{T_{co} - T_{ci}} = \dfrac{130 - 110}{85 - 15} = 0.286$

Temp ratio $P = \dfrac{T_{co} - T_{ci}}{T_{hi} - T_{ci}} = \dfrac{85 - 15}{130 - 15} = 0.608$

From data book $F = 0.97$ (approx.)

↳ correction factor

$$A = \frac{Q}{U \times (LMTD) \times F} = \frac{193.44 \times 10^3}{275 \times 66.91 \times 0.97}$$

$$= 10.838\ m^2$$